

PC5203 Advanced Solid State Physics

Weeks 12-13, due Thursday 22 Apr 2021

[main concepts to cover: Berry phase, Berry curvature, superconductivity, BCS theory.]

1. Consider the phonon Hall model Hamiltonian given as $H = \frac{1}{2}(p - Au)^2 + \frac{1}{2}u^T Ku$, where u is a column vector of displacements relative to the equilibrium positions, p is the conjugate momentum, K is a symmetric force constant matrix, and A is an antisymmetric matrix. T stands for matrix transpose. p^2 means $p^T p$ for brevity.
 - a. Derive the classical Hamilton equations of motion from the Hamiltonian given.
 - b. Assuming periodicity of a two-dimensional crystal lattice, we can rewrite the vector u as $u_{\mathbf{l},j}$, where $\mathbf{l} = (l_1, l_2)$ is pair of integers such that real space lattice vector is given as $\mathbf{R}_{\mathbf{l}} = l_1 \mathbf{a}_1 + l_2 \mathbf{a}_2$. The unit cell is described by the two $\mathbf{a}$ vectors. The index j specifies the degrees of freedom in a unit cell, e.g., for graphene, $j = 1$ to 6 as there are two atoms per unit cell, and each atom can move in x, y, z directions. Because of lattice periodicity, both matrix A and K is a function of the difference in $\mathbf{l}$. Derive the equation of motion in $\mathbf{q}$ space, that is, the Fourier transform in index $\mathbf{l}$, with the usual convention. Give the relation of the Dynamic matrix D to the real space K , and define the $\mathbf{q}$ space A .
 - c. Assuming all the atoms move in one single frequency ω in the normal mode, derive the eigenvalue problem $H_{\text{eff}}\Psi = \omega\Psi$ (that is, give the expression for the effective Hamiltonian and the definition of Ψ is a column vector consisting of position u and velocity v in $\mathbf{q}$ space).
 - d. From the definition of Berry connection $d\gamma = -\text{Im} \bar{\Psi} d\Psi$, derive the Berry curvature formula (for each fixed $\mathbf{q}$)

$$\Omega_n = -\text{Im} \sum_{m \neq n} \frac{\bar{\Psi}_n \partial_x H_{\text{eff}} \Psi_m \bar{\Psi}_m \partial_y H_{\text{eff}} \Psi_n}{(\omega_n - \omega_m)^2} - (x \leftrightarrow y)$$

Here we sum over m excluding the case of $m = n$, for a fixed n . And $\bar{\Psi} = \Psi^\dagger \begin{bmatrix} D & 0 \\ 0 & I \end{bmatrix}$, $D = D(\mathbf{q})$ is the usual dynamic matrix, and I is the identity matrix (all in space of site index j). We assume the modes are normalized according to $\bar{\Psi} \Psi = 1$. The index m or n labelled the eigenmodes, the partial derivative is with respect to q_x for ∂_x and similar for y component. The second subtraction term is obtained by swapping x with y.

- e. Discussing. What happen if the phonon dispersion happens to have only one band (i.e., one degree per unit cell with $j = 1$ and the matrix H_{eff} is 1 by 1 like in a 1D chain)? What happen if two bands touch each other with a degeneracy?

2. The free energy in a conventional superconductor is $F = \int \frac{1}{2} m \mathbf{v}^2 n d^3 \mathbf{r} + \int \frac{1}{2\mu_0} \mathbf{B}^2 d^3 \mathbf{r} + \text{const.}$

- a. By minimizing the free energy, together with the definition of current, $\mathbf{j} = (-e)n\mathbf{v}$, and the Ampere's law, $\nabla \times \mathbf{B} = \mu_0 \mathbf{j}$, derive the London equation

$$\mathbf{j} = -\frac{ne^2}{m} \mathbf{A}$$

Here the vector potential $\mathbf{A}$ is transverse, i.e., $\nabla \cdot \mathbf{A} = 0$. $\mathbf{B} = \nabla \times \mathbf{A}$. We treat the electron density n as a constant.

- b. Why the London equation (in another form) implies the Meissner effect, that is, the magnetic induction $\mathbf{B}$ is zero inside a superconductor body?

3. The (spinless, one-dimensional) BCS ground state is postulated to be

$$|\text{BCS}\rangle = \prod_{p>0} (u_p + v_p c_p^\dagger c_{-p}^\dagger) |0\rangle$$

here $|0\rangle$ is the electron vacuum state, u_p and v_p are complex numbers.

- a. Determine the condition needed on the coefficient u and v if $\langle \text{BCS} | \text{BCS} \rangle = 1$, that is, the many-particle state is normalized to one.
- b. Compute the expectation value of BCS Hamiltonian in the ground state, $E_g = \langle \text{BCS} | \hat{H}^{\text{BCS}} | \text{BCS} \rangle$, here

$$\hat{H}^{\text{BCS}} = \sum_p (\epsilon_p - \mu) c_p^\dagger c_p + \sum_{p,q>0} V_{pq} c_p^\dagger c_{-p}^\dagger c_{-q} c_q$$

(This is actually a K-miltonian, $\hat{K} = \hat{H} - \mu \hat{N}$, because we subtracted a chemical potential term). We assume the interaction matrix V is symmetric with respect to the index p and q .

- c. By minimizing the ground state energy, E_g in part b, derive the solution of u and v to be, for each wavevector p , $u^2 = \frac{1}{2} (1 + \frac{\xi}{E})$, $v^2 = \frac{1}{2} (1 - \frac{\xi}{E})$, $\xi = \epsilon - \mu$, and the quasi-particle energy $E = \sqrt{\xi^2 + \Delta^2}$, and $\Delta_p = -\sum_{q>0} V_{pq} u_q v_q$. This is the gap equation in BCS theory.